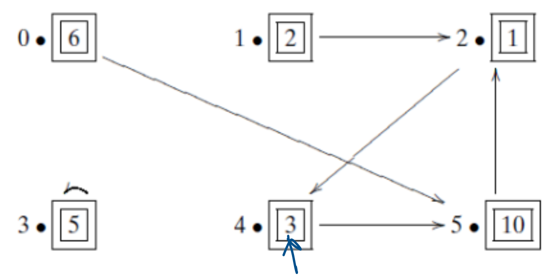
3.1 A Leap from Classical to Quantum

Sunday, October 5, 2025 6:10 PM

3.1.1 Classical Deterministic Systems

Example 3.1.1: Consider the simple system described by a graph of 6 vertices together with 27 marbles. Imagine the identical marbles as being placed on the vertices of a graph.

$$X = \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array}\begin{bmatrix} 6 \\ 2 \\ 1 \\ 5 \\ 3 \\ 10 \end{bmatrix}$$



$$M = \begin{array}{c|cccccc} i \backslash j & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 5 & 1 & 0 & 0 & 0 & 1 & 0 \end{array}$$

The movement of the marbles in the graph require a deterministic system. In other words, each marble must move to exactly one place. The idea is that if an arrow exists from vertex i to vertex j , then in one time click, all the marbles on vertex i will shift to vertex j .
This graph is easy to store in a computer as a Boolean adjacency matrix, M

$$M = \begin{array}{c} \\ \mathbf{0} \\ \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \\ \mathbf{4} \\ \mathbf{5} \end{array}\begin{array}{c} \begin{array}{cccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array},$$

In general, any simple directed graph with n vertices can be represented by an n-by-n matrix M having entries as

$$M[i, j] = 1 \text{ if and only if there is an edge from vertex } j \text{ to vertex } i.$$
$$= 1 \text{ if and only if there is a path of length 1 from vertex } j \text{ to vertex } i.$$
$$M^k[i, j] = 1 \text{ if and only if there is a path of length } k \text{ from vertex } j \text{ to vertex } i.$$

Example 3.1.2: Using the dynamics M given in Example 3.1.1, determine what the state of the system would be if you start with the state $X = [6\ 2\ 1\ 5\ 3\ 10]^T$.

vertex: 0 1 2 3 4 5

vertices.

$$M_{6\times 6} X_{6\times 1} = M = \begin{array}{c} \\ \mathbf{0} \\ \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \\ \mathbf{4} \\ \mathbf{5} \end{array}\begin{array}{c} \begin{array}{cccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}, \begin{bmatrix} 6 \\ 2 \\ 1 \\ 5 \\ 3 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \\ 5 \\ 1 \\ \ldots \end{bmatrix} \begin{array}{c} \to 0 \\ \to 1 \\ \to 2 \\ \to 3 \\ \to 4 \\ \to 5 \end{array}$$

6x6 6x1

$$\begin{array}{c|c}
3 & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \\
4 & \\
5 &
\end{array}
\begin{bmatrix} 5 \\ 3 \\ 10 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 9 \end{bmatrix} \begin{matrix} \to 3 \\ \to 4 \\ \to 5 \end{matrix}$$

$\underline{}$ 27

Example 3.1.3: For the matrix M given in above example, calculate M^2, M^3, and M^6. If all the marbles start at vertex 2, where will all the marbles end up after 6 time steps?

$$M = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array},$$

$M^2 = MM$, $M^3 = M^2 M$

$M^6 = M^3 M^3$

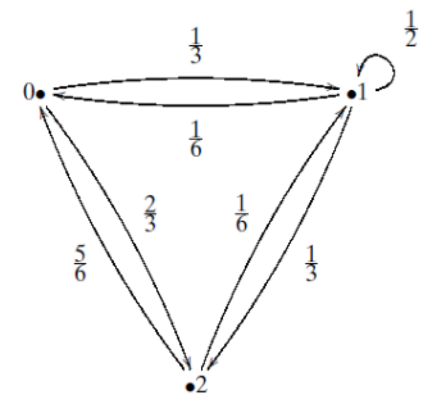
Summarizing up, we have learned the following:
- The states of a system correspond to column vectors (state vectors).
- The dynamics of a system correspond to matrices.
- To progress from one state to another in one time step, one must multiply the state vector by a matrix.
- Multiple step dynamics are obtained via matrix multiplication.

5.1.2 Probabilistic Systems

In quantum mechanics, our knowledge of a physical state is inherently uncertain, states change with probabilistic laws. In otherwards, the system's evolution are given by describing how states transition from one to another with a certain likelihood.

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

Example 3.1.4: Consider the following directed graph, write the adjacency matrix



$$M = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ \\ \end{array} \begin{array}{c} \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{bmatrix} 0 & \frac{1}{6} & \frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix} \\ \begin{matrix} 1 & 1 & 1 \end{matrix} \end{array} \begin{matrix} = 1 \\ = 1 \\ = 1 \end{matrix}$$

Definition 3.1.1:(**Doubly stochastic matrix**) A matrix with all real entries between 0 and 1 where the sums of the rows and the sums of the columns are all 1

Example 3.1.5: Consider the state $X = \left[\frac{1}{6}\ \frac{1}{6}\ \frac{2}{3}\right]^T$, that expresses an indeterminacy about the position of a marble. Find the probability of the marble's location after moving using the adjacency matrix in example 3.1.4.

$$Mx = \begin{bmatrix} 0 & \frac{1}{6} & \frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 7/12 \\ 1/4 \\ 1/6 \end{bmatrix}$$

5.1.3 Quantum Systems

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$c = a + bi$

$|c|^2 = a^2 + b^2$

- Quantum mechanics works with complex numbers, c.
- A weight of a complex number c such that $|c|^2$ is a real number.
- Unlike real probabilities, if c_1 and c_2 be two complex numbers with associated squares of modulus $|c_1|^2$ and $|c_2|^2$. $|c_1 + c_2|^2$ need not be bigger than $|c_1|^2$ and it also does not need to be bigger than $|c_2|^2$.
- **Interference** is an important concepts in quantum theory, where **quantum probability amplitudes** (not probabilities themselves) **combine or cancel each other out** "*complex numbers may cancel each other out when added*"

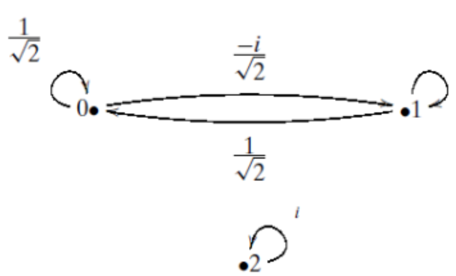
Example 3.1.6: Consider $c_1 = 5 + 3i$ and $c_2 = -3 - 2i$

$$|c_1 + c_2|^2 = |2+i|^2 = (2+i)(2-i) = 5$$

$$|c_1|^2 = 25 + 9 = 34 \qquad |c_2|^2 = 9 + 4 = 13$$

$$|c_1 + c_2|^2 < |c_1|^2 \;\&\; |c_2|^2 .$$

Example 3.1.7: Consider the following directed graph, write the adjacency matrix U, is U a unitary matrix?



$$U = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & i \end{bmatrix} \end{array}$$

Remark: Unitary matrices are related to doubly stochastic matrices, the modulus squared of the all the complex entries in U forms a doubly stochastic matrix

Example 3.1.8: Find $|U[i,j]|^2$

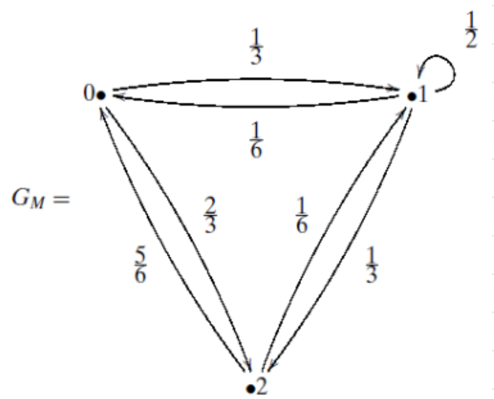
$$\begin{array}{c} \\ 0 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \end{array} \qquad \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

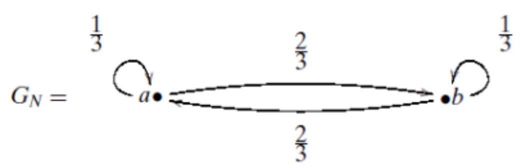
$$
U = \begin{array}{c} 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & i \end{bmatrix} \Longrightarrow \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

$$
UU^{\dagger} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & i \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} & 0 \\ 0 & 0 & -i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

3.1.4 Assembling Systems

Example 3.1.9: Consider two different marbles. The red marble follows the probabilities of the graph G_M and the blue marble that follows the graph G_N





Their corresponding adjacency matrices are:

$$
M = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0 & \frac{1}{6} & \frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix} \end{array} \qquad N = \begin{array}{c} \begin{array}{cc} a & b \end{array} \\ \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \end{array}.
$$

How does a state for a two-marble system look? How does a state for a two-marble system look?

$$X = \begin{array}{c} 0a \\ 0b \\ 1a \\ 1b \\ 2a \\ 2b \end{array} \begin{bmatrix} \frac{1}{18} \\ 0 \\ \frac{2}{18} \\ \frac{1}{3} \\ 0 \\ \frac{1}{2} \end{bmatrix},$$

$$M \otimes N = \begin{array}{c} \\ \mathbf{0}a \\ \mathbf{1} \\ \mathbf{2} \end{array} \begin{array}{c} \begin{array}{ccc} \qquad\mathbf{0}\qquad & \qquad\mathbf{1}\qquad & \qquad\mathbf{2}\qquad \end{array} \\ \begin{bmatrix} 0\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{6}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{5}{6}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \\ \frac{1}{3}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{2}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{6}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \\ \frac{2}{3}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{3}\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & 0\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \end{bmatrix} \end{array}$$

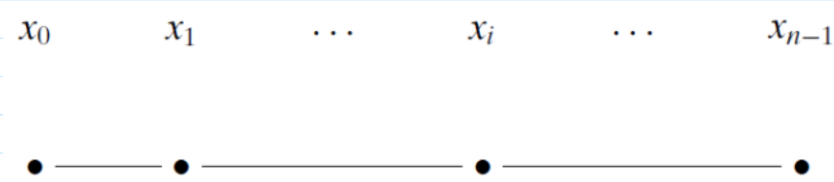
$$= \begin{array}{c} \\ \mathbf{0}a \\ \mathbf{0}b \\ \mathbf{1}a \\ \mathbf{1}b \\ \mathbf{2}a \\ \mathbf{2}b \end{array} \begin{array}{c} \begin{array}{cccccc} \mathbf{0}a & \mathbf{0}b & \mathbf{1}a & \mathbf{1}b & \mathbf{2}a & \mathbf{2}b \end{array} \\ \begin{bmatrix} 0 & 0 & \frac{1}{18} & \frac{2}{18} & \frac{5}{18} & \frac{10}{18} \\ 0 & 0 & \frac{2}{18} & \frac{1}{18} & \frac{10}{18} & \frac{5}{18} \\ \frac{1}{9} & \frac{2}{9} & \frac{1}{6} & \frac{2}{6} & \frac{1}{18} & \frac{2}{18} \\ \frac{2}{9} & \frac{1}{9} & \frac{2}{6} & \frac{1}{6} & \frac{2}{18} & \frac{1}{18} \\ \frac{2}{9} & \frac{4}{9} & \frac{1}{9} & \frac{2}{9} & 0 & 0 \\ \frac{4}{9} & \frac{2}{9} & \frac{2}{9} & \frac{1}{9} & 0 & 0 \end{bmatrix} \end{array}.$$

3.2 Basic Quantum Theory

Monday, October 6, 2025 2:25 PM

3.2.1 Quantum states

Consider a subatomic particle on a line, which can be detected at one the equally distant point



Thus, the current sate of the particle is a complex column vector $\left[c_0, c_1, \ldots, c_{n-1}\right]^T$. The particle being at the point x_i shall be denoted as $|x_i\rangle$, this notation called **ket**.
These basic states $|x_i\rangle$ at the point x_i are formed the standard basis of $\mathbb{C}^n$

$$|x_0\rangle \longrightarrow [1, 0, \ldots, 0]^T$$

$$|x_1\rangle \longrightarrow [0, 1, \ldots, 0]^T$$

$$\vdots$$

$$|x_{n-1}\rangle \longrightarrow [0, 0, \ldots, 1]^T$$

$\langle x_0 | \psi \rangle = c_0$

Definition 3.2.1: An arbitrary state, which we shall denote as $|\psi\rangle$, will be a linear combination of $|x_0\rangle, |x_1\rangle, \ldots, |x_{n-1}\rangle$, by suitable complex weights, $c_0, c_1, \ldots, c_{n-1}$, known as **complex amplitudes.**

$$|\psi\rangle = c_0|x_0\rangle + c_1|x_1\rangle + \cdots + c_{n-1}|x_{n-1}\rangle$$

We say that the state $|\psi\rangle$ is a **superposition** of the basic states. $|\psi\rangle$ represents the particle as being simultaneously in all $\{x_0, x_1, \ldots, x_{n-1}\}$ locations.

- The complex numbers $c_0, c_1, \ldots, c_{n-1}$ tell us precisely which superposition the particle is currently in.
- The norm square of the complex number c_i divided by the norm squared of $|\psi\rangle$ will tell us the probability that, after observing the particle, we will detect it at the point x_i:

$$p(x_i) = \frac{|c_i|^2}{||\psi\rangle|^2} = \frac{|c_i|^2}{\sum_j |c_j|^2}.$$

Example 3.2.1: Let us assume that the particle can only be at the four points and its state is given by

$$|\psi\rangle = (-3 - i)|x_0\rangle - 2i|x_1\rangle + i|x_2\rangle + 2|x_3\rangle$$

Calculate the probability that the particle can be found at position x_1

$|c_1|$

$\langle x_1 | \psi \rangle = -2i$

$$P(x_1) = \frac{(-2i)(2i) = -4i^2 = 4}{16 + 4 + 1 + 4} = \frac{4}{19}$$

$$p(x_i) = \frac{(-2i)(2i)}{16+4+1+4} = \frac{}{19} \quad \#$$

Properties of ket:

1. Addition:

$$|\psi\rangle = c_0|x_0\rangle + c_1|x_1\rangle + \cdots + c_{n-1}|x_{n-1}\rangle = [c_0, c_1, \ldots, c_{n-1}]^T$$

and

$$|\psi'\rangle = c'_0|x_0\rangle + c'_1|x_1\rangle + \cdots + c'_{n-1}|x_{n-1}\rangle = [c'_0, c'_1, \ldots, c'_{n-1}]^T,$$

then

$$\begin{aligned}|\psi\rangle + |\psi'\rangle &= (c_0 + c'_0)|x_0\rangle + (c_1 + c'_1)|x_1\rangle + \cdots + (c_{n-1} + c'_{n-1})|x_{n-1}\rangle \\ &= [c_0 + c'_0, c_1 + c'_1, \ldots, c_{n-1} + c'_{n-1}]^T.\end{aligned}$$

2. Scaler multiplication: for $c \in \mathbb{C}$

$$c|\psi\rangle = cc_0|x_0\rangle + cc_1|x_1\rangle + \cdots + cc_{n-1}|x_{n-1}\rangle = [cc_0, cc_1, \ldots, cc_{n-1}]^T.$$

$$\begin{aligned}|\psi\rangle + |\psi\rangle = 2|\psi\rangle &= [c_0 + c_0, c_1 + c_1, \ldots, c_j + c_j, \ldots, c_{n-1} + c_{n-1}]^T \\ &= [2c_0, 2c_1, \ldots, 2c_j, \ldots, 2c_{n-1}]^T.\end{aligned}$$

3. The ket $\underline{\underline{c|\psi\rangle}}$ describes the same physical system as $\underline{\underline{|\psi\rangle}}$.

$$p(x_i) = \frac{\overset{1}{\cancel{|c|^2}}|c_i|^2}{\underset{1}{\cancel{|c|^2}}\left(|c_0|^2 + |c_1|^2 + \cdots + |c_{n-1}|^2\right)}$$

Example 3.2.2: Determine whether the following vector describe the same quantum states.

$$|\psi_1\rangle = \begin{bmatrix} 1+i \\ i \end{bmatrix} \quad \text{and} \quad |\psi_2\rangle = \begin{bmatrix} 2+4i \\ 3i-1 \end{bmatrix}$$

$$|\psi_1\rangle = c|\psi_2\rangle$$

$$c = \frac{|\psi_1\rangle}{|\psi_2\rangle}$$

$$|\psi_2\rangle = c|\psi_1\rangle \Rightarrow c = \frac{|\psi_2\rangle}{|\psi_1\rangle}$$

$$\frac{1-i}{1-i} \times \frac{2+4i}{1+i} \overset{?}{=} \frac{3i-1}{i} \times \frac{-i}{-i}$$

$$\frac{6+2i}{} = 3+i = \frac{3+i}{}$$

$$\frac{6+2i}{2} = 3+i = \frac{3+i}{1}$$

Note: As we can multiply (or divide) a ket by any (complex) number and still have are presentation of the same physical state $|\psi\rangle$:

Normalization: $$\frac{|\psi\rangle}{||\psi\rangle|}$$

Example 3.2.3: Find the length of the following vector $[2+3i\,, 1-2i]^T$ and then normalize it.

$$||\psi\rangle| = \sqrt{\langle\psi|\psi\rangle}$$

$$\langle\psi|\psi\rangle = \begin{bmatrix} 2-3i & 1+2i \end{bmatrix} \begin{bmatrix} 2+3i \\ 1-2i \end{bmatrix}$$

$$= \sqrt{13+5} = \sqrt{18} = 3\sqrt{2}$$

$$\frac{|\psi\rangle}{||\psi\rangle|} = \left[\frac{2+3i}{3\sqrt{2}}\,,\ \frac{1-2i}{3\sqrt{2}}\right]^T$$

Spin Property: spin **up** $|\uparrow\rangle$ and spin **down** $|\downarrow\rangle$. The state in the supperposition

$$|\psi\rangle = c_0|\uparrow\rangle + c_1|\downarrow\rangle.$$

Example 3.2.4: Consider a particle whose spin is described by the ket

$$|\psi\rangle = (3-4i)|\uparrow\rangle + (7+2i)|\downarrow\rangle.$$

Find the amplitude of detecting the spin of the particle in the down direction.

$$\langle\downarrow|\psi\rangle = (3-4i)\overbrace{\langle\downarrow|\uparrow\rangle}^{=0} + (7+2i)\overbrace{\langle\downarrow|\downarrow\rangle}^{=1}$$

$$= 7+2i$$

Transition amplitudes: is the inner product of the state space, which in turn will enable us to determine how likely the state of the system before a specific measurement (start state), will change to another (end state), after measurement has been carried out.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\langle 0|\psi\rangle = \alpha$$

$$\underset{\text{Start state}}{|\psi\rangle} \xrightarrow{\langle\psi|\psi'\rangle} \underset{\text{End state}}{|\psi'\rangle}.$$

$\langle\psi|$ is called **bra** state defined as $\langle\psi| = |\psi\rangle^\dagger$

$$\langle\psi'|\psi\rangle$$

Note: The transition amplitude between two states may be zero. In fact, that when the two states are orthogonal to one another

Example 3.2.5: Compute the amplitude of the transition from $|\psi\rangle = \frac{\sqrt{2}}{2}[1, i]^T$ to $|\phi\rangle = \frac{\sqrt{2}}{2}[i, -1]^T$

$$\langle \varphi | \psi \rangle = \frac{\sqrt{2}}{2}[-i, -1] \cdot \frac{\sqrt{2}}{2}\begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$= \frac{1}{2}(-i - i) = -i$$